

A NOVEL BLOCK-CHAIN BASED ENERGY ANTI-THEFT SYSTEM ADVANCED WITH GRAPH THEORY AND MACHINE LEARNING ALGORITHM FOR REAL-TIME ANOMALY DETECTION

ABSTRACT :

Electricity theft causes substantial economic losses on a global scale comprising an estimated global loss of over \$96 billion. It's a prevailing issue in India itself specially in states like Bihar having the malpractices like Meter Tampering, Line Tapping, Political Corruption. This paper presents a novel, cost-effective anti-theft system based on the use of blockchain, graph theory and Machine Learning and micro-controller based meters to ensure a transparent, inviolable and effective electricity distribution immune to all the above mentioned malpractices. Blockchain secures data integrity, Graph theory enables real-time anomaly detection and ML provides the scope for predictive maintenance and accuracy. The solution is tailored with cost-effectiveness, innovation, feasibility, impact, efficiency and scalability whether in Rural or Urban Areas.

HIGHLIGHTS

1. “DECENTRALIZED, BLOCKCHAIN POWERED SMART GRID”

- The system utilizes blockchain to create a decentralized tamper-proof ledger to record energy consumption data ensuring transparency, inviolable security, record tamper-proofing and accountability.

2. AI-POWERED SELF TRAINING ANOMALY DETECTION

- The system uses advanced AI algorithms like Isolation Forest and LSTM, for the real-time detection of unusual consumption patterns and adapts itself through self-retraining providing an intelligent and highly accurate real-time monitoring

3. SCALABLE, EFFICIENT AND TACTICAL DESIGN

- Designed with cost-effective and low energy consumption and microcontrollers and optimized communication protocols considering the cost effectiveness of the implementation in both rural and urban areas making it adaptive and scalable for different areas and large scale implementation.

4. MULTI LAYER SECURITY AND LIVE TAMPER ALERTS

- Incorporates energy meter hardware tamper alerts , Meter & SUBSTATION Unique ID , AES encryption based data transfer and multi-factor inter-monitoring between different monitoring levels to ensure security at both physical and digital grounds.

5. GRAPH THEORY BASED OPTIMIZATION TO COMPLIMENT AI ALGORITHM EFFICIENCY. PURPOSING LOUAVAIN ALGORITHM

1.Introduction

Electricity theft is a prevalent issue worldwide, leading to significant financial losses and energy inefficiency . Besides these, electric blackouts are among the major consequences of this theft. Conventional monitoring systems lack the required sensitivity and adaptability to address existing challenges due to many vulnerabilities. The critical vulnerabilities such as inaccuracies , Inability to detect distributed energy thefts. i.e. small energy thefts from different areas in a vicinity , Meter Tampering , Political influence to manipulate the energy consumption data at ground level, Direct Line Tapping , Corrupted staff at substations selling the electricity. This paper proposes a robust inviolable system for energy distribution that ensures secure data integrity, transparency , real-time anomaly detection , predictive maintenance , tamper proofing of consumption records using blockchain technology , microcontroller based meters , Graph theory and Machine Learning. The paper follows that via detailed analysis of the impact and future possibilities .

2. PROPOSED SYSTEM ARCHITECTURE

The architecture operates across three main levels : Household Meter Units, Local Substations and Central Station. 1. The micro-controller based meters monitor the home usage and report to substation with AES encryption . 2. Local substations : Keeping the track of the consumption records in a distributed tamper-resistant ledger and creating blocks, and also monitoring energy thefts at sub-region level. 3. Central Station : All the advanced anomaly detections occurs here , This component also ensures occurrence of tampering at substations (Thwarting fake substations and manipulation at that level). First Stage includes each meter having a Unique ID thwarting the possibility of

fraudulent meter connections . Second Stage eliminates the possibility of direct line tapping. Third Stage : Eliminates the possibility of all the other mentioned malpractices by monitoring the fraudulent reporting at substation level. Even Political corruption at any level can do nothing because of blockchain records and inter-monitoring of different stages.

THE MEDIUM OF DATA TRANSFER BETWEEN METERS AND SUBSTATION HAS BEEN MENTIONED LATER IN THE RESEARCH

2.1 LOCAL MONITORING

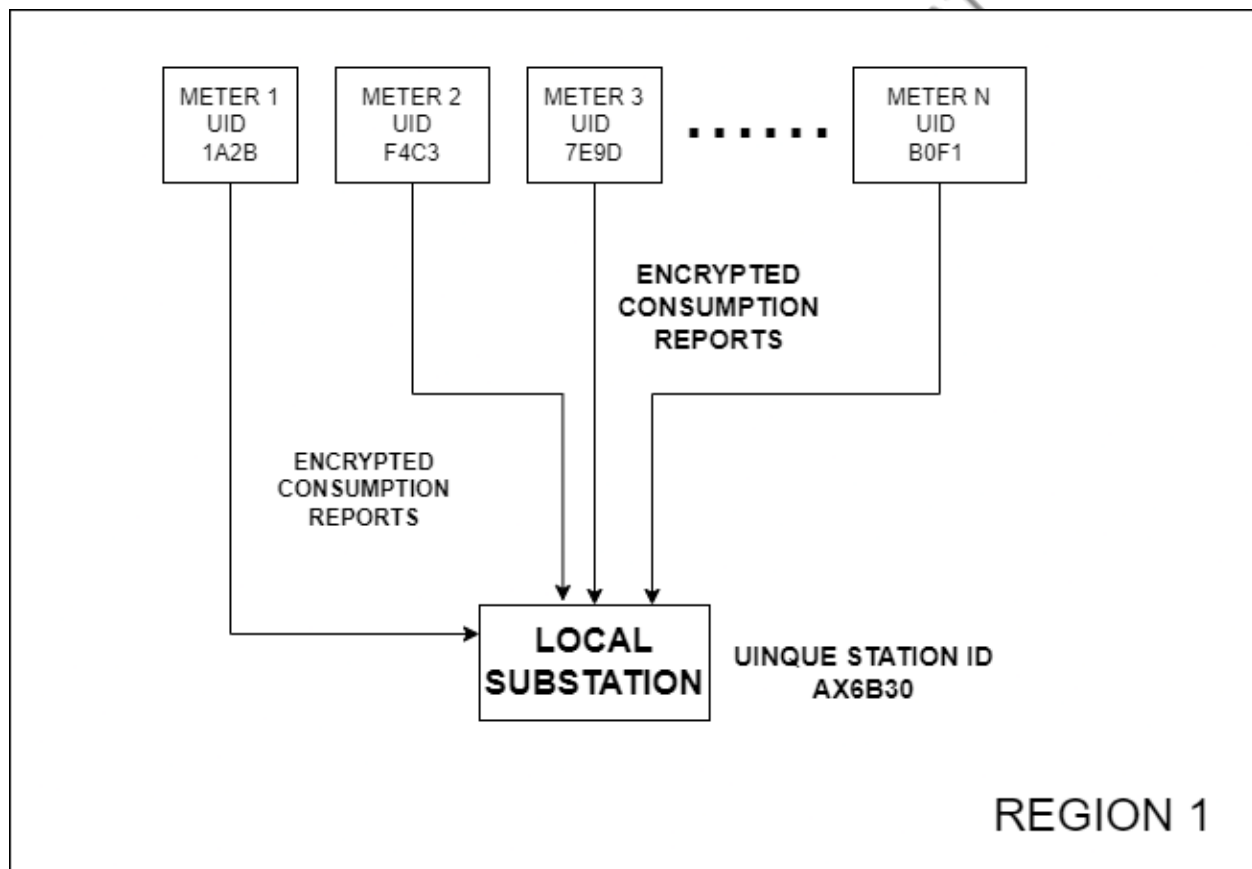


IMAGE 2.1. : LOCAL MONITORING

WHENEVER A CONSUMER METER ATTEMPTS TO CONNECT TO THE SUBSTATION , AN AUTHENTICATION IS MADE BEFORE THE CONNECTION . MICRO-CONTROLLER SENDS A PING TO THE LOCAL

SUBSTATION (A CONNECT REQUEST) USING AES ENCRYPTION. EACH PING MESSAGE WOULD BE AS FOLLOWS.

```
{  
  "unique_id": "METER_ID",  
  "nonce": "UNIQUE_NONCE",  
  "timestamp": "TIMING ACCORDING TO RESPECTIVE TIME ZONE",  
  "encrypted_payload": "ON OR OFF",  
  "signature": "base64_encoded_signature" // Digital signature  
  generated using the device's private key  
}
```

PING MESSAGE

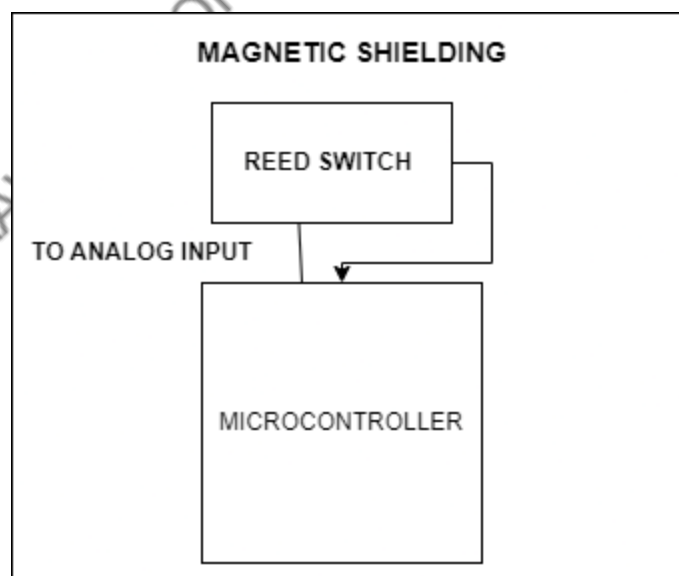
The substation verifies the ping via digital signature , device id and nonce and timestamp (to check if the ping was made whilst the permissible time period to avoid replay attacks) . Once it is verified the meter's connection to powerline is accepted and acknowledgment packet is sent to the meter as confirmation.

2.1.1 HOUSEHOLD METER UNITS

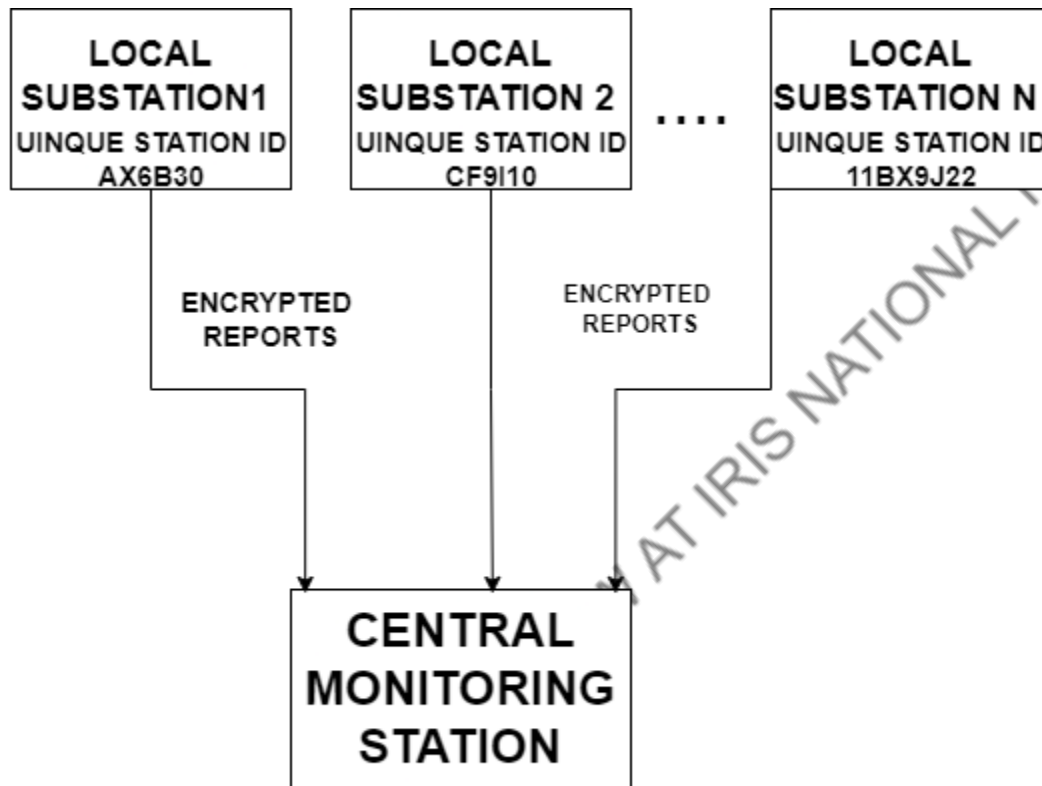
Each household meter has cost-effective microcontrollers like STM8 series microcontrollers or Arduino Nano with current sensors to monitor consumption. Apart from that a tamper protection is also there near the microcontroller unit. Microcontroller PCB is placed in a steel unit called tamper protection unit.

2.1.1.1 METER TAMPER PROTECTION UNIT

- Steel has been used to enclose the tamper protection unit because steel acts as magnetic shield . This ensures that reed switch doesn't get falsely activated due to magnetic fields created due to electric currents in the meter's components. Hence, avoid false tampering alerts.
- **If the steel box is unscrewed in attempt to manipulate the microprocessor** : This will cause the reed switch to activate which will save a tamper log in microprocessor memory and send the same to the substation.



2.2 LOCAL SUBSTATIONS



**CENTRAL MONITORING
(STATE POWER STATION)**

FUNCTIONS AT SUBSTATION LEVEL :

1. MAINTAINING **BLOCKCHAIN** LEDGER : Local Blockchains of the consumptions are maintained and added to a distributed Ledger of the overall consumption of the Major Region . Each Substation serves as a node allowing for decentralized collection of data.
2. LOCAL ANOMALY DETECTION USING GRAPH THEORY : LOUVAIN METHOD CLUSTERS HOUSEHOLDS WITH SIMILAR CONSUMPTION, INDICATING OUTLIERS INDICATIVE OF THEFT. APART FROM THAT

Z-SCORE ANALYSIS IS ALSO USED TO DETECT “NODES” (HOUSES IN THIS CONTEXT- NOTE : POWER NODE NOT BLOCKCHAIN NODE) WITH ABNORMALLY HIGH OR LOW CONSUMPTION.

3. REPORTING TO CENTRAL MONITORING STATION.

CHALLENGES AND SOLUTIONS

1. HIGH-END HARDWARE REQUIREMENTS TO GENERATE PROOF OF WORK FOR BLOCK VERIFICATION :

SOLUTION :

APPLICATION OF PRACTICAL BYZANTINE FAULT TOLERANCE AS CONSENSUS MECHANISM

THE MENTIONED MECHANISM HAS BEEN USED TO VERIFY THE PERMISSION FOR A METER'S DATA TO BE ADDED TO A BLOCK IN THE BLOCKCHAIN . (BLOCKCHAIN CREATION)

IT OVERCOMES THE PROBLEM OF COMPUTATION EFFORT BECAUSE IT SIMPLY DOESN'T REQUIRE IT.

2. POSSIBILITY OF CORRUPTION AT GROUND LEVEL : IT'S NOT POSSIBLE BECAUSE THERE'S A DISTRIBUTED LEDGER MAINTAINING ALL THE CONSUMPTION RECORDS ACROSS THE SUBSTATIONS AND CENTRAL STATIONS. IT WON'T BE PRACTICAL TO MANIPULATE EACH TO TAMPER FEW RECORDS.

3. COST-EFFECTIVENESS : AS SOLUTION TO CHALLENGE 1 SUGGESTS THERE'S NO REQUIREMENT OF COMPUTATION EFFORT THEREFORE NO HIGH-END HARDWARE REQUIRED.

2.3 CENTRAL MONITORING SYSTEM

The central system monitors the substations. It employs ML and advanced analytics for that sake.

- ISOLATED FOREST ALGORITHM : Data logs form the dataset that is taken in purpose by isolated forest algorithm to detect outliers eliminating even the minute possibility of fraud at substation.

PROBLEM ELIMINATED :

The possibility was there that one could've created a fake substation and added it to the central monitoring by generating unique station and private key using political contacts and personal contacts with central monitoring station personnel. (Targeting the loopholes of PBFT consensus mechanism). This possibility has been eliminated using data-set analysis done by the isolated forest algorithm. Refer to algorithms section for explanation of exact working.

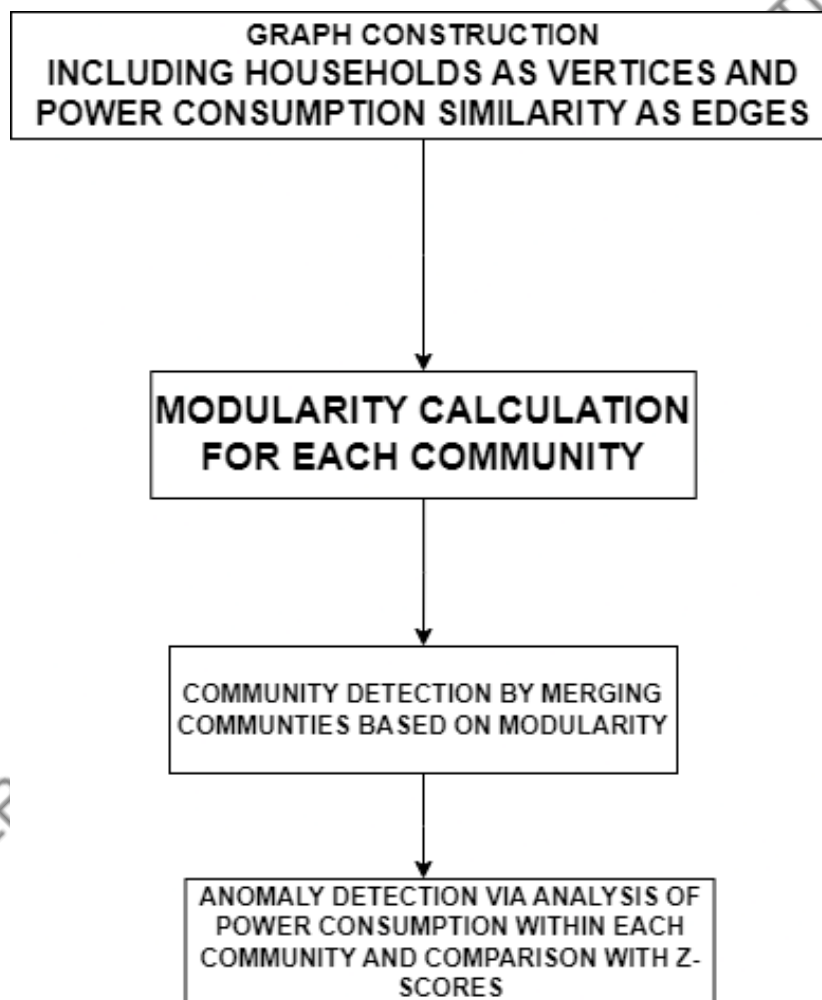
- K-MEANS CLUSTERING : THIS improves efficiency in dataset processing. This clustering categorizes substations based on energy usage, reducing the computational complexity at Central Station Level and improving accuracy.

PROBLEM ELIMINATED :

HUGE DATASET MAY DECREASE THE ANALYSIS ACCURACY UPON REAL-TIME ANALYSIS. THIS SOLVES THAT ISSUE BY STRATEGIC PARTITION OF DATA. ELABORATED IN ALGORITHMS SECTION.

2.4 ALGORITHMS AND MATHEMATICAL JUSTIFICATION

- **LOUVAIN ALGORITHM** : Tactically uses pre-computed (**HISTORIC DATA**) **Z-SCORES** of houses in a community to identify the unusually high or low consumption of households in a neighbourhood in comparison to other households in the same locality.



GRAPH CONSTRUCTION

Create a graph $G = (V, E)$ V represents set of Households AND E the set of edges representing the similarity in power consumptions among the houses. Here cosine similarity metric has been used to assign weights to these edges.

$$K = \left(\frac{1}{2m} \right) * \sum_{(i,j) \in E} (A_{ij} - \frac{k_i * k_j}{2m})$$

m : Total no. of edges

A_{ij} : Adjacency matrix element

k_i : Degree of a node

K : Modularity of that community

E : Set of Edges

NOW, It's an optimization problem to maximize K for that sake Louvain algo is used repeatedly to combine communities resulting in k_{max} . Once these communities are established, Z-scores are used to pinpoint exact location of anomalies in the community.

THE COMPUTATIONAL IMPLEMENTAION OF THE SAME HAS BEEN SHARED ON MY GITHUB PAGE MENTIONED IN REFERENCE .

➤ **Z-SCORE SETUP**

1. COLLECTION OF HOUSEHOLD CONSUMPTION DATA

$$P=[P_1,P_2,P_3,\dots,P_n]$$

2. Mean (μ) Calculation

3. SD CALCULATION (σ)

4. Z SCORE FOR EACH HOUSE = $(P_i - \mu) / \sigma$

P_i : Power consumption of individual house

➤ **ISOLATION-FOREST ALGORITHM**

The fundament it works upon is that unusual observation can be separated from the majority of the data points with fewer splits.

SAMEER SAAWAN (FOR THE VIEW AT IRIS INTERNATIONAL FAIR)

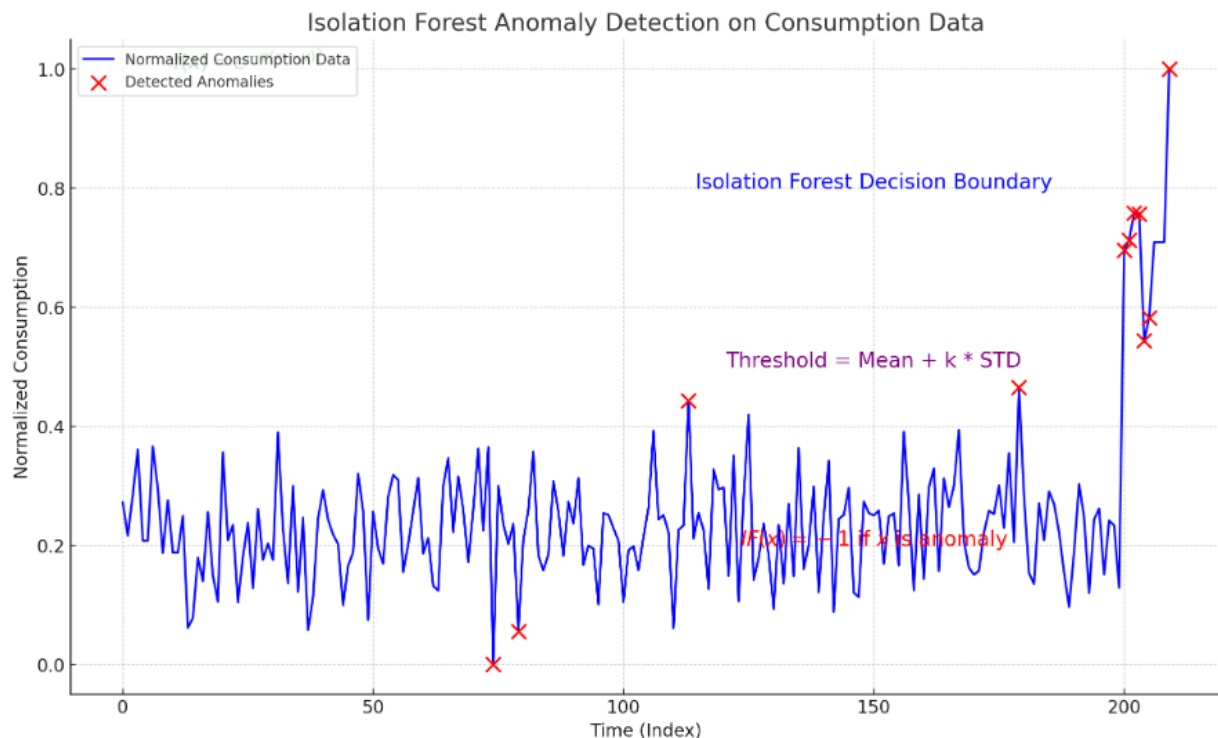
```
[Start]
|
[Input Data]
|
[Random Sampling]
|
[Construct Isolation Trees]
|
+--> [For each subset:]
|   |
|   [Recursive Splitting]
|   |
|   +--> [Choose feature f]
|   |
|   +--> [Select split value s]
|   |
|   +--> [Split observations]
|   |   |
|   |   [Observations < s] --> [Left Branch]
|   |   |
|   |   [Observations ≥ s] --> [Right Branch]
|   |
|   +--> [Repeat until stopping criteria met]
|   |
|   [Tree Formation] --> [Create Isolation Tree (iTree)]
|
[Path Length Calculation]
|
[Anomaly Score Computation]
|
[Aggregate Scores from Multiple Trees]
|
[Anomaly Classification]
|
[End]
```

- LET $L(X)$ BE THE LENGTH FROM THE CORRESPONDING LEAF TO THE ROOT NODE. (X IS AN OBSERVATION)
- LET $U(N)$ BE THE AVERAGE PATH LENGTH OF UNSUCCESSFUL SEARCHES IN BINARY TREE. $U(N) = 2 * \ln(N - 1) + 0.5772$
- LET $F(X)$ BE THE ANOMALY SCORE FUNCTION. $F(X) = 2^{-(L(X)/U(N))}$.
- $F(X)$ MEAN FOR DIFFERENT X IS THE FINAL FOREST THRESHOLD. IF IT EXCEEDS THE SYSTEMATIC POWER CONSUMPTION THRESHOLD THEN IT IS CLASSIFIED AS ANOMALY.

SOFTWARE IMPLEMENTATION OF THE ALGORITHM

The following are the specifications of the program mentioned on my github page :

- ADVANCED HYPER PARAMETER TUNING – USING GRIDSEARCHCV
- USES THE LEDGER DATA TO MAKE DATASET



ANOMALY DETECTION SIMULATION WITH RANDOM PARAMETERS
(MATPLOT SIMULATION)

2.5 DATA-TRANSFER MEDIUM

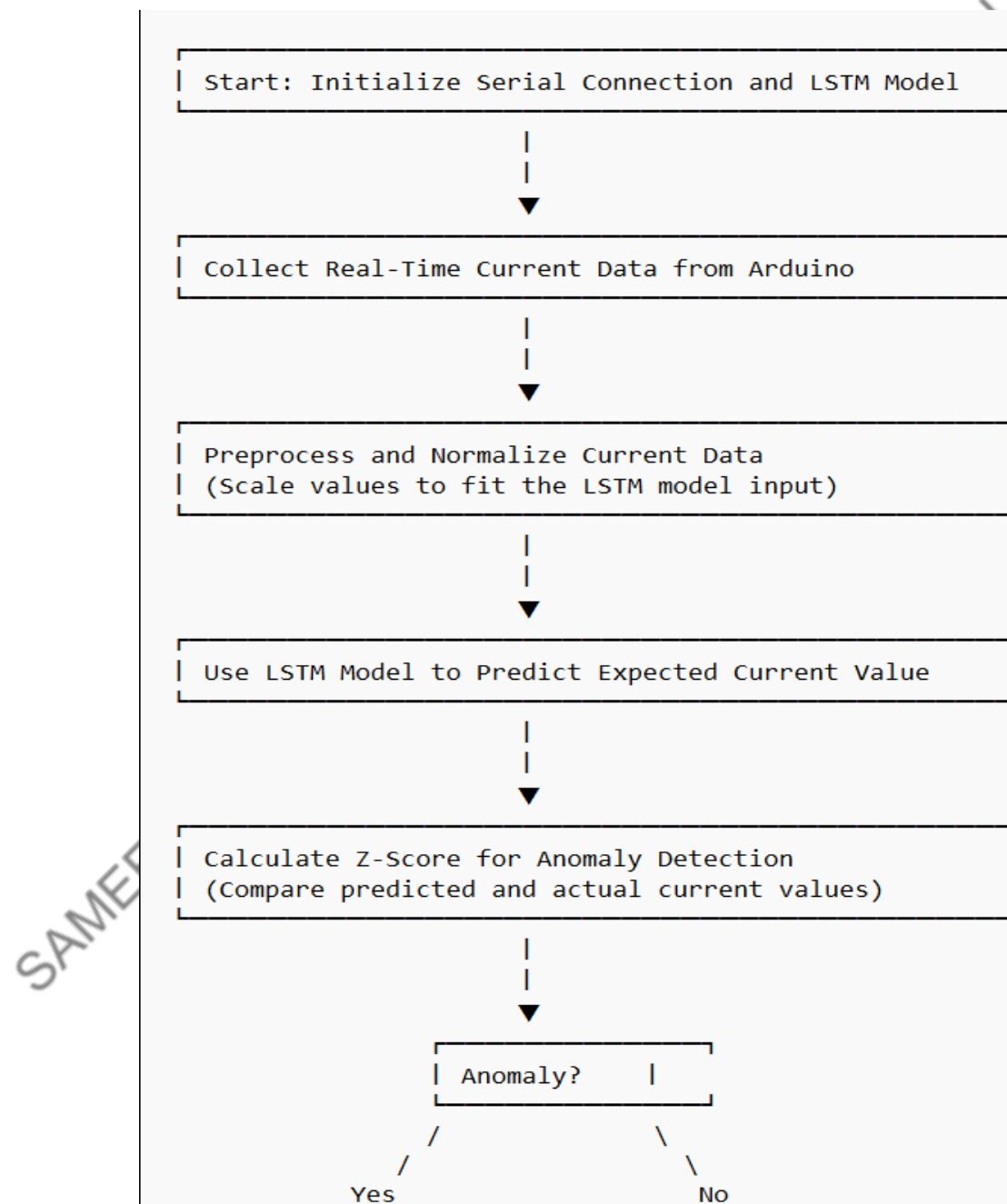
Data transfer mediums are tailored as per the area of implementation.

- RURAL AREAS : LoRaWAN (LONG RANGE WIDE AREA NETWORK) , RF COMMUNICATION
 - URBAN AREAS : THE ABOVE MENTIONED & WIFI
- RF COMMUNICATION
- COST-EFFECTIVE
 - OPERATES AT LOW POWER CONSUMPTION

HARDWARE : Low cost transmission devices like Zigbee modules , Power Source: Solar Panels and backup batteries .

Specific Transmission frequency : A specific transmission frequency is chosen as per the legal decisions.

2.6 PREDICTIVE MAINTENANCE



Log Anomaly in Ledger (with timestamp and details)	Continue Logging Data to Ledger
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SOFTWARE IMPLEMENTATION – Program has been shared on my github page mentioned as reference.

Program Salient feature : Use of Tensorflow library to implement LSTM neural network for time series prediction. It is to be trained to learn current draw patterns of different components to be able to predict unusual current drains indicating malfunctioning of electrical components. Providing a solution for predictive maintenance , Yet. It is to be trained.

4. ANALYSIS AND RESULTS

THE SOLUTION OFFERS A DEDICATED ROBUST,SCALABLE AND COST-EFFICIENT SOLUTION FOR ENERGY THEFT, AS PER THE SIMULATED DATASET , THE SYSTEM IS CAPABLE TO ACHIEVE **EFFICIENCY UPTO 96% (REFERENCE : MY GITHUB PAGE)**

- IMPACT : ALMOST EVERY POSSIBLE FOR THEFT HAS BEEN CONSIDERED . THE INTEGRATION OF ML AND GRAPH THEORY BASED ALGORITHMS PROMISE A RELIABLE AND HIGHLY ACCURATE MONITORING.

5. CONCLUSION

THE SYSTEM HAS THE CAPABILITY TO IMPROVE THE ECONOMICAL AND TECHINICAL EFFICIENCY BY A SUBSTANTIAL FIGURE WITH NEGLIGIBLE HUMAN OVERSIGHT AND LEAST INFRASTRUCTURE

REFERENCE :

MY GITHUB Page :

https://github.com/SAMEER-MLPROJECT/AI_ANTI-THEFT_SYSTEM